COM661 Full Stack Strategies and Development

SEMESTER 1, 2026-2027

Assignment 1

"Individual Full Stack Application Development – Back End"

Over the course of the module, we construct the ***Biz Directory*** - a sample full stack application built around a database containing information on a range of businesses and a collection of user-contributed reviews. This assignment tests your understanding of this material by having you build the **back-end** of your own full stack application, based around a dataset of your choice. You can obtain your dataset from any source, but you should modify its structure (if required) to provide the most appropriate fit for your planned functionality.

Your application can be in any field but must be driven by a back-end **Python Flask** API that interfaces with a **MongoDB** database and must provide opportunities to demonstrate all of the main database operations (create, retrieve, update, delete).

The primary focus should be on **usability** and **functionality**. The application should fulfil a well-defined need and should provide a range of API endpoints that could potentially support different front-end applications.

Learning Outcomes Assessed

This assignment addresses all module learning outcomes.

Marking Criteria

Submissions will be marked according to the following criteria

- Database structure and appropriateness to the application
- Range and complexity of database queries
- Provision of API endpoints as RESTful resources with appropriate use of HTTP verbs and status codes
- Use of individual user accounts and appropriate authentication
- Evidence of automated API testing
- Code structure and presentation

- Provision of appropriate documentation

This assignment is worth **50%** of the total mark for the module

Effort

This assignment has a total notional workload of **20 hours**.

Submission

The deadline for submission is 12pm (Noon) on **Friday 6th November 2026**. All work should be submitted to the **CW1 Submission** link in the **Assessment** area of Blackboard and should consist of the following elements:

- A Zip file containing all code files for the back-end API.
- A Zip file containing a JSON file for each MongoDB collection in the database. JSON files should be generated by running the **mongoexport** command for each collection.
- A video submission that (i) introduces your application area and dataset, (ii) demonstrates your API in use with **Postman**, and (iii) provides a walkthrough of your code. The maximum length of the video is **5 minutes**.

  Please note the following penalty for exceeding workload which will be applied for videos that exceed the maximum length:
  Ref: https://www.ulster.ac.uk/learningenhancement/cqe/assessment-code-of-practice/penalties-for-exceeding-assessment-limits

  "Where submitted work exceeds the agreed assessment limit, a margin of up to +10% of the work limit will be allowed without any penalty of mark deduction.

  If the work submitted is significantly in excess of the specified limit (+10%), there is no expectation that staff will assess the piece beyond the limit or provide feedback on work beyond this point. Markers will indicate the point at which the limit is reached and where they have stopped marking. A mark will be awarded only for the content submitted up to this point. No additional deduction or penalty will be applied to the overall mark awarded. The student is self-penalising as work will not be considered/marked."

- Separate PDF files containing
  - a complete code listing of the back-end API
  - a summary of all API endpoints

    - a printout of the result of running an automated collection of tests on the API through Postman
    - a printout of the API documentation generated through Postman
- A completed self-evaluation sheet on which you provide a critical appraisal of your submission against a range of marking criteria.

A video demonstrating the preparation of a submission pack for the sample ***Biz Directory API*** application will be made available on Blackboard.

VERY IMPORTANT: The application code and video demonstration are crucial pieces of evidence in assessing your contribution to the work. If either of these is not present, your potential for gaining high marks will be severely affected.

Use of AI

You are permitted to use AI tools such as GitHub CoPilot in this assignment. However, it is very important that you can be seen to fully understand the application structure and code. The code walkthrough section of the video should focus on demonstrating in-depth understanding of the most complex aspects of code and the Design and Development document should discuss the algorithms and data structures used, giving clear justification for their selection and identifying any potential alternatives. **Where a Chat Agent has been used, you should include the chat history in the Design and Development document, showing clearly your prompts and the Agent's responses.**

Marking Scheme

You should be able to judge the level of your submission by reference to the following criteria. Note that the ***Biz Directory*** example referred to here, relates to the state of the application at the end of **Practical BE08.** Please note that a detailed rubric and marking guide is provided at the end of this document.

Classification	Criteria
1st Class (70%+)	The application will be fit-for-purpose and usable for the intended task. Functionality will be significantly in excess of that demonstrated by the ***Biz Directory API*** example with database operations beyond the

	standard CRUD functionality. The application will be robust with appropriate validation and authentication and will be free from bugs.
2:1 (60-69%)	The application will have a level of functionality above that of the ***Biz Directory API*** example but there may be minor functionality deficiencies that prevent it from being truly fit for the intended purpose. The application should be free from major bugs that prevent elements of functionality from running properly.
2:2 (50-59%)	The application will be similar to the ***Biz Directory API*** example in terms of working functionality and data structure but will suffer from bugs or performance issues that prevent it from being a usable application. There will be at least some evidence that the developer has attempted to extend the basic functionality.
3rd (40-49%)	The application will be heavily influenced by the ***Biz Directory API*** example, with extended re-use of code without modification or addition. There will have been no visible attempt to enhance the functionality and there may be significant bugs that prevent it from working properly.
Fail (<40%)	The application will suffer from major bugs that prevent it from being properly examined. There will have been no attempt to enhance functionality from the ***Biz Directory API*** example

Feedback

An **optional interim submission** during Week 4 will consist of a short (one page max) summary of your data set, application domain and implementation plan. This submission is for **feedback only** (i.e., no marks are awarded for this) and feedback on this will be provided through Blackboard within one week.

Individual feedback on each final submission will be provided in written form under the categories presented in the marking criteria above.

Feedback on final submissions will be returned within 20 working days after submission in accordance with Faculty policy.

Plagiarism Statement

By making a submission, you will be deemed to have made the following declaration of ownership.

"I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I understand the importance of academic integrity and have read and understood the University's General Regulation: Student Academic Integrity and the Academic Misconduct Procedure. I understand that I must not upload my work before, during or after submission to any unapproved plagiarism detectors or answer sharing platforms, or equivalent, and that only University-approved platforms should be used."

Coursework 1 – ASSESSMENT RUBRIC/MARKING PROFORMA

Criteria (100%)	80-100% (High 1st)	70-79% (1st)	60-69% (2.1)	50-59% (2.2)	40-49% (3rd)	<40% (Fail)
Choice of Dataset 15%	Comprehensive dataset as for 70-79% but additional merit (e.g., from an original or unusual source, containing non text/numeric data, multiple sub-documents, etc.).	Comprehensive dataset with a large collection, good range of types, and populated sub-documents. If the original dataset has been modified to add additional fields or types, these should be explained.	All desired features present, but dataset is limited in size, not fully populated, or with only one or two data types.	Usable dataset but missing some desirable feature (e.g., sub-document collections, range of different data types).	Basic dataset which largely follows the structure of that provided in the demonstration application	Insufficient data to demonstrate full functionality or data from the module material used.
Database Functionality 25%	As for 70-79% but a comprehensive set of queries demonstrated, exceeding the complexity demonstrated in the module material. Evidence of additional research, e.g. query types not covered in the module material.	As for 60-69% with evidence of complex queries such as multi-valued search, aggregation pipeline, geo-location queries, etc.	As for 50-59% but with use of searching and matching within CRUD operations.	All CRUD operations implemented on top-level and sub-document collections.	All CRUD operations implemented on top-level collections.	Basic CRUD operations incomplete.
API Structure and Design 25%	As for 70-79%, but a level of functionality far in advance of the demonstration application and immediately usable for the stated purpose. Evidence of additional research such as Python constructs not covered in the module material or application-specific detail. Where AI has been used, it has been employed responsibly, enriching student's understanding.	As for 60-69% with comprehensive back-end provision at a level greater than that provided by the demonstration application. Appropriate use of modular Blueprints in API design. Where AI tools have been used, these are acknowledged and the student demonstrates command of the resulting codebase and provides rationale in their own words.	As for 50-59% plus additional endpoints to support increased functionality (e.g., pagination, filtering, etc.) AI-assisted sections, if any, are acknowledged.	As for 40-49% plus support for full set of CRUD operations on documents and sub-documents. AI-assisted sections, if any, are mostly credited. Where an agent has been used, the chat history has been provided.	API design is in the RESTful style with basic CRUD operations supported on top-level documents Some AI-assisted elements may be present but may not be fully justified or acknowledged. Where an agent has been used, the chat history has not been provided.	Less than the basic CRUD operations supported. Code or discussion appears AI-generated and is unexplained or unacknowledged.
Usability 25%	As for 70-79% but evidence of additional polishing – e.g. validation includes type and range checking, provision of informative exception messages, etc.	As for 60-69% but with a comprehensive range of functionality that could potentially support multiple front-end applications.	As for 50-59% but all user input checked and appropriate HTTP status codes returned.	Some additional functionality over the demonstration application but limited by incomplete validation or inappropriate use of HTTP Status Codes.	Basic functionality on a par with the demonstration application.	Some functionality, but not all CRUD operations supported and/or essential error trapping not present.
Submission Package 10%	Evidence of exceptional attention to detail in testing, documentation and/or video presentation.	All elements submitted to a good standard with no major deficiency (e.g., coverage of testing and documentation, lack of detail in the video).	All elements submitted but one exhibits some deficiency.	All elements submitted, but multiple deficiencies (e.g., coverage of POSTMAN testing and documentation, lack of detail in the video).	One supporting document missing.	Multiple supporting documents missing.